SOC Incident Report

End-to-End Intrusion Detection and Analysis Lab

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Chapter 1

Executive Summary

This project simulates a real-world cyber attack scenario to analyze attacker behavior and detection techniques. The lab includes reconnaissance, exploitation, and web-based attacks using industry-standard tools. The objective is to identify indicators of compromise (IOCs) and understand how a Security Operations Center (SOC) analyst detects and responds to threats.

Chapter 2

Environment Setup

Component	Details
Attacker Machine	Kali Linux
Target Machine	Metasploitable2
Monitoring Tool	Wireshark

Tools Used

- Nmap (Network Scanning)
- Wireshark (Packet Analysis)
- Metasploit (Exploitation Framework)
- Burp Suite (Web Security Testing)

Chapter 3

Attack Timeline

Time	Activity
10:00	Network scanning using Nmap
10:05	Service and version detection
10:10	Exploitation using Metasploit
10:15	Web login attack using Burp Suite
10:20	Traffic analysis using Wireshark

Chapter 4

Reconnaissance Analysis (Nmap)

The attacker performed network scanning to identify open ports and services.

Findings

- Open Ports:
  - 21/tcp (FTP)
  - 22/tcp (SSH)
  - 80/tcp (HTTP)
- Service Versions:
  - vsftpd 2.3.4 (Vulnerable)
  - Apache HTTP Server
  - OpenSSH

Security Insight

Outdated services increase the risk of exploitation.

Chapter 5

Exploitation Analysis (Metasploit)

The attacker exploited a known vulnerability in the FTP service.

Details

- Exploit Used: vsftpd_234_backdoor
- Target Port: 21 (FTP)
- Result: Successful command shell access

Impact

- Unauthorized remote access gained
- System integrity compromised
- Potential lateral movement

Chapter 6

Web Attack Analysis (Burp Suite)

The attacker performed a login attack by intercepting and modifying HTTP requests.

Observations

- HTTP POST requests intercepted
- Login credentials modified
- Multiple login attempts observed

Indicators

- Repeated login attempts
- Parameter tampering
- Brute-force behavior

Chapter 7

Network Traffic Analysis (Wireshark)

Observations

Port Scanning Activity

Multiple SYN packets were detected across various ports, indicating scanning behavior.

Exploitation Traffic

FTP communication showed abnormal behavior and suspicious responses.

Web Attack Traffic

Multiple HTTP POST requests were observed indicating repeated login attempts.

Chapter 8

Indicators of Compromise (IOCs)

- High number of SYN packets
- Suspicious FTP activity
- Repeated HTTP login attempts
- Unauthorized shell access

Chapter 9

Risk Assessment

Threat	**Severity**
Port Scanning	Medium
Service Exploitation	High
Brute Force Attack	High

Chapter 10

Mitigation Recommendations

- Disable unused ports and services
- Patch vulnerable software
- Implement firewall rules
- Deploy Intrusion Detection Systems (IDS)
- Enable account lockout policies
- Continuous log monitoring

Chapter 11

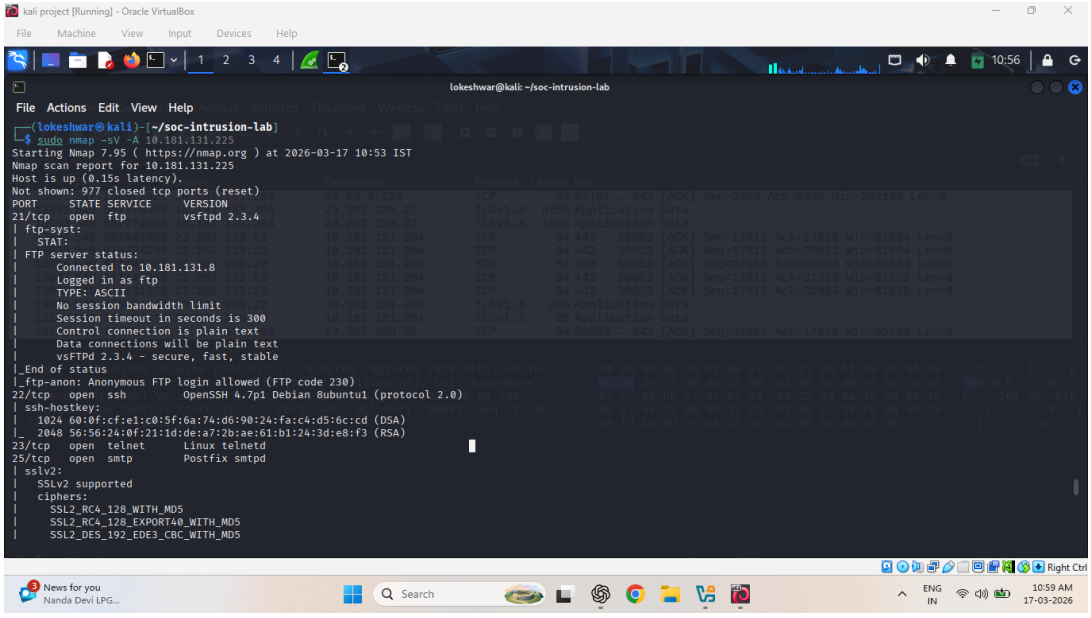
Conclusion

This lab demonstrates the complete attack lifecycle from reconnaissance to exploitation and web attacks. It highlights how attackers operate and how their activities can be detected using network monitoring tools. This project strengthens practical skills required for a SOC Analyst role.

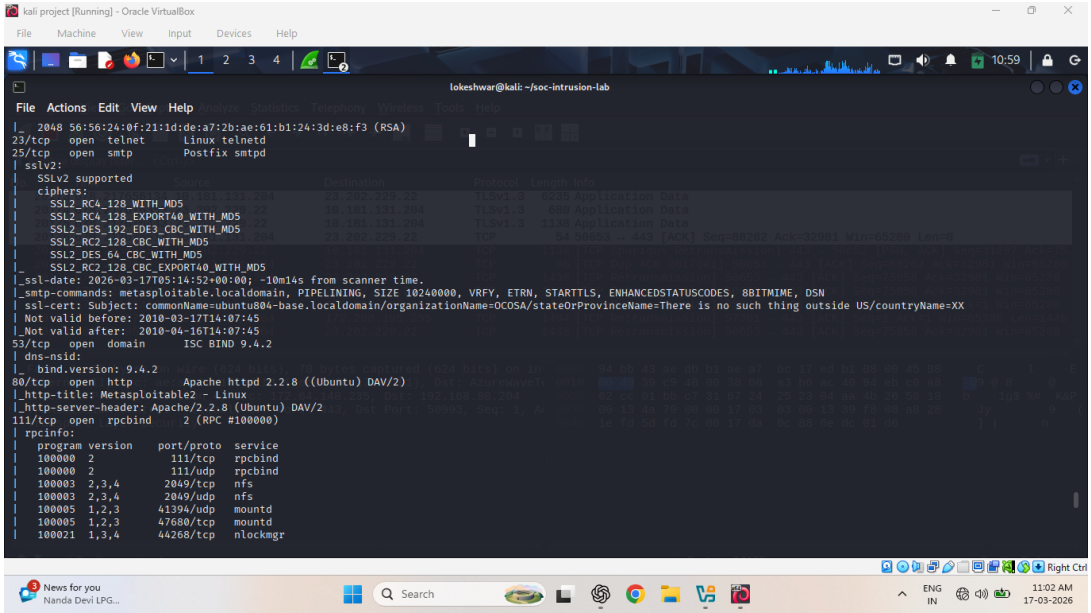
Chapter 12

Screenshots

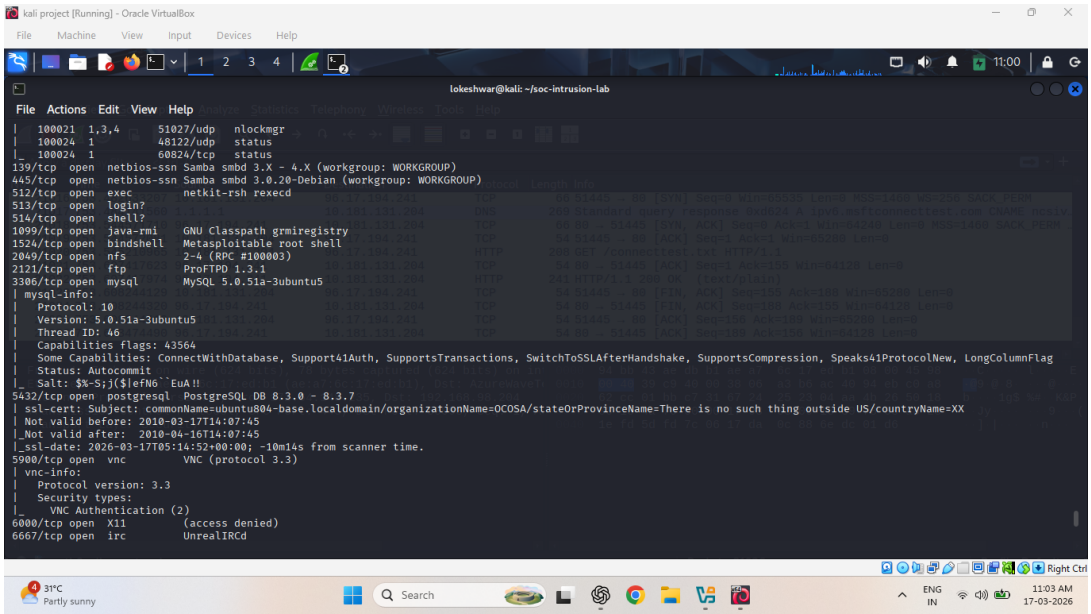
Nmap Scan



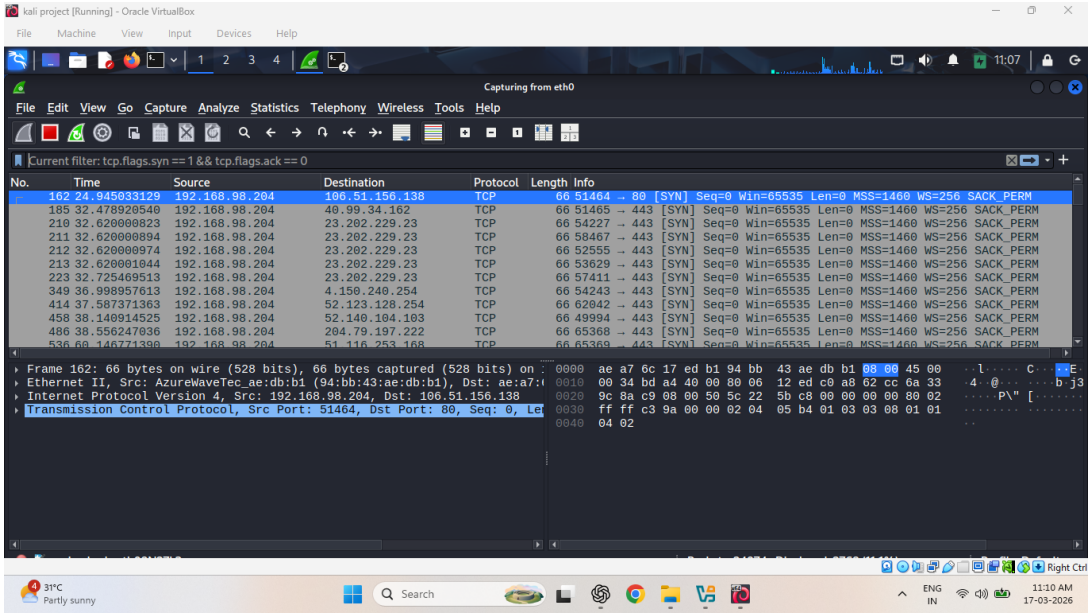
Nmap Scan



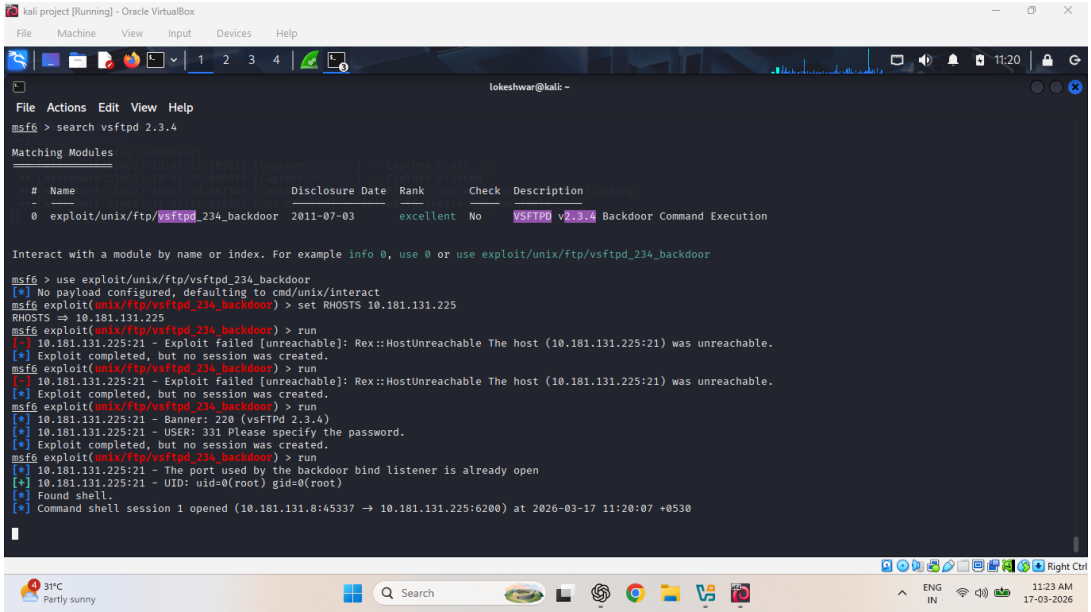
Nmap Scan



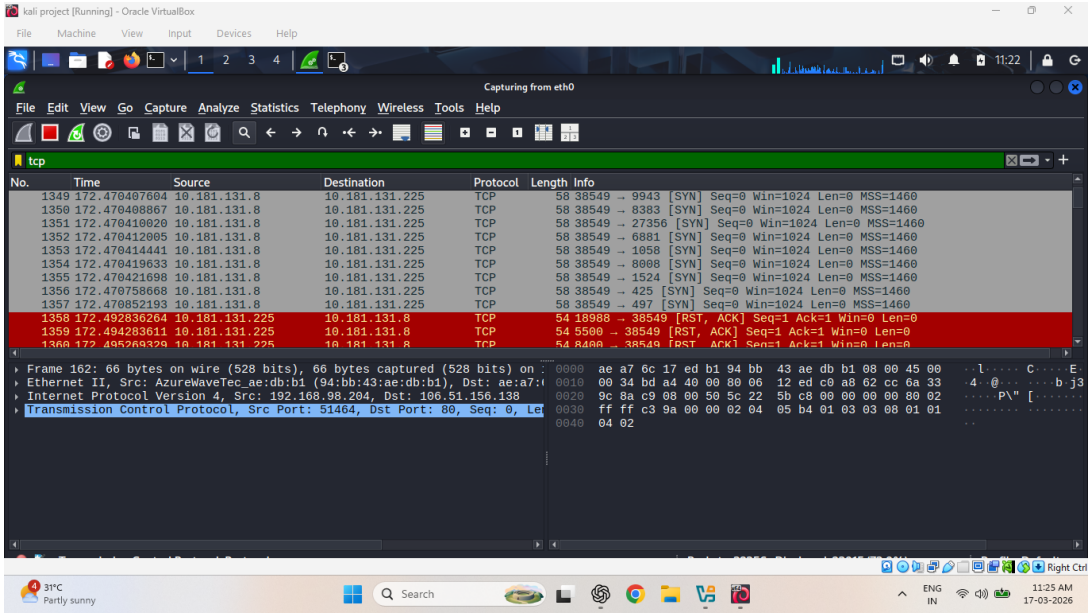
SYN Scan Detection



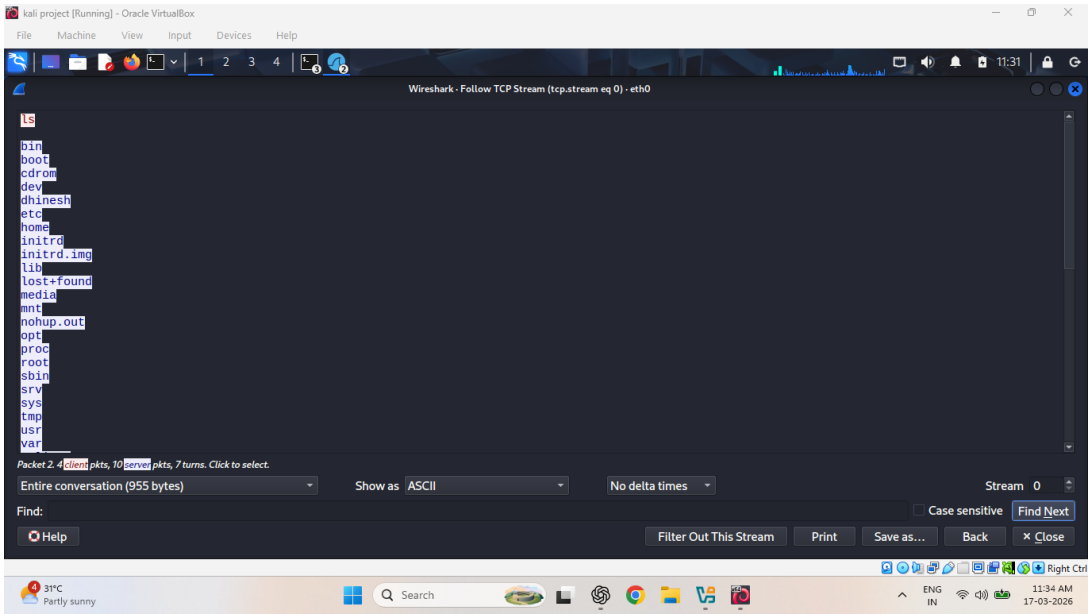
Metasploit Exploitation



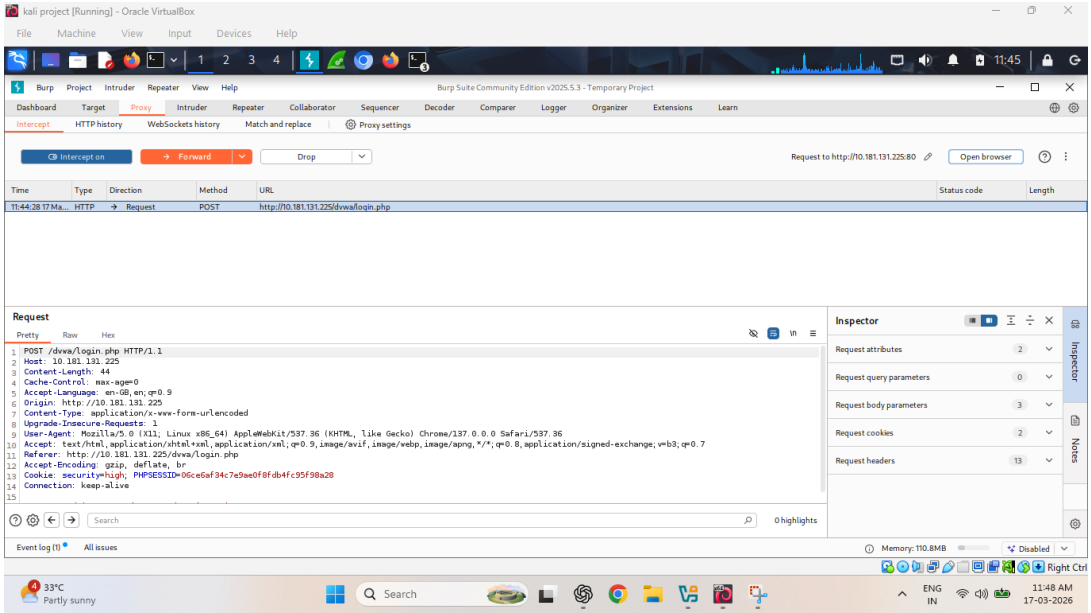
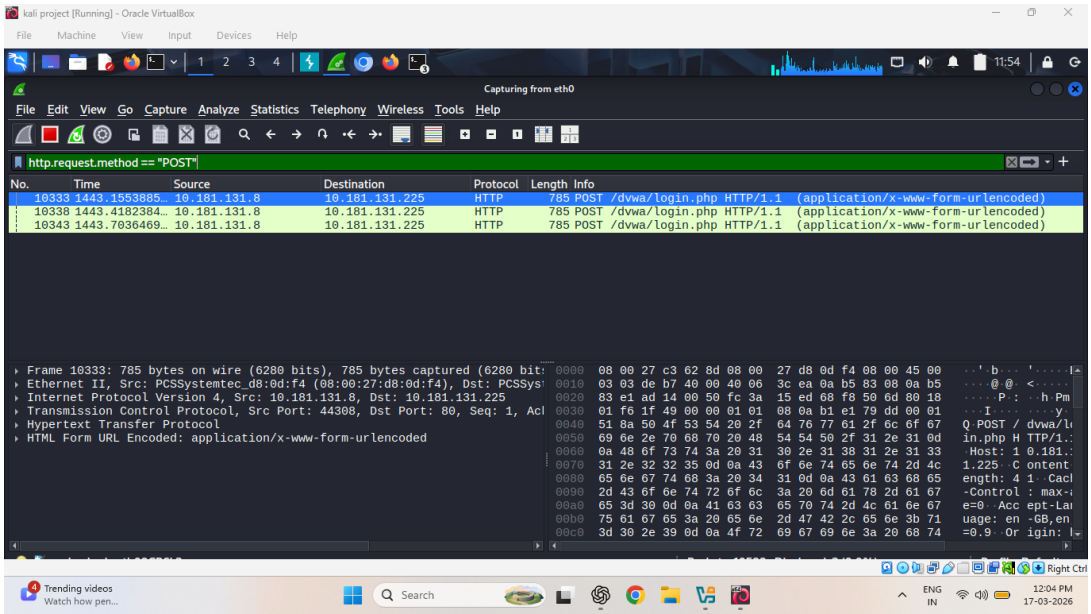
Metasploit Exploitation



Metasploit Exploitation



Burp Suite Interception



HTTP Attack Traffic